Saravana Raja S

BACKEND & CLOUD ENGINEER · IOT / REALTIME / INTEGRATIONS

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01 PROFILE

Backend and cloud engineer with 2+ years building **cloud-connected IoT platforms**, real-time telemetry ingestion, and healthcare integrations. I work on the unglamorous layer: MQTT and WebSocket gateways, device state, time-series analytics, HL7 interfaces, and the Linux/Docker/AWS deployment underneath.

02 EXPERIENCE

Lead Engineer · Mudivili Technologies, Chennai — 02/2025 – 09/2026

Node.js · FastAPI · MQTT · WebSockets · PostgreSQL · TimescaleDB · HL7 · AWS

- Built a **real-time telemetry gateway** (FastAPI, MQTT, WebSockets) handling 100+ devices publishing every five seconds (~20 msg/s), streaming live state to 1,000+ connected clients.
- Added in-memory device-state caching, removing per-message database reads for live state; used TimescaleDB continuous aggregates for low-latency historical dashboards.
- Designed a **Laboratory Information System interface** in Python over HL7/TCP, integrating medical analyzers with the cloud and removing manual result transcription.
- Built secure Node.js/Express services with JWT authentication, RBAC, and REST APIs shared by mobile, frontend, and embedded teams.
- Deployed containerized services on AWS EC2 with Docker, Nginx, SSL/TLS, Linux services, and long-running MQTT processes; improved reliability through unit tests and protocol validation.

Cloud Engineer 1 · Hanu Software Solutions, Greater Noida — 09/2022 – 08/2023

C# · .NET · Microsoft Azure · Azure SDK · Scheduled jobs

- Developed an internal .NET application to monitor Azure resource usage and surface cost-optimization insights.
- Integrated the Azure SDK for resource details, usage metrics, and cost data; added rule-based detection of underutilized VMs and idle resources.
- Built scheduled background jobs for data collection and reporting to internal teams.

Customer Support Associate · Sutherland Global Services, Chennai — 08/2018 – 12/2020

- Supported retail customers with order issues and product troubleshooting — the origin of a bias toward clear, usable error paths.

03 SELECTED PROJECT

Support & Bug Triage Agent · Amazon Bedrock AgentCore

- Guardrailed tool-calling agent on Nova Pro with deterministic required-field collection, session isolation, and DynamoDB ticket creation through AgentCore Gateway and Lambda.
- Infrastructure as code with CloudFormation; repeatable LLM-as-a-judge evaluation pipeline using Bedrock Model Evaluations for correctness and completeness.

github.com/saravana-raja/bedrock-agentcore-support-triage

04 TECHNICAL SKILLS

LANGUAGES

JavaScript (ES6+), Python, C#, SQL

BACKEND

Node.js, Express.js, FastAPI, REST APIs, JWT / RBAC

MESSAGING & PROTOCOLS

MQTT, WebSockets, TCP/IP, HL7 v2

DATA

PostgreSQL, TimescaleDB, MySQL, Redis, SQL

CLOUD & PLATFORM

AWS (EC2), Microsoft Azure, Docker, Docker Compose, Linux, Nginx

TOOLS

Git, GitHub, Postman, VS Code

05 CERTIFICATIONS

AWS Certified Solutions Architect — Associate
Amazon Web Services · 2026

AWS AI/ML Scholars — Agentic AI
Udacity · 06-10/2026

Microsoft Certified: Azure Developer Associate
Microsoft · 2022

Junior Full-Stack Java Developer Bootcamp
Generation India · 2021-22

06 EDUCATION

B.E. Mechanical Engineering
Infant Jesus College of Engineering and Technology, Thoothukudi
08/2013 – 05/2017

07 LANGUAGES

English fluent · Hindi basic · Tamil native

● OPEN TO BACKEND / CLOUD / IOT ROLES · CHENNAI OR REMOTE